Experiment-1:

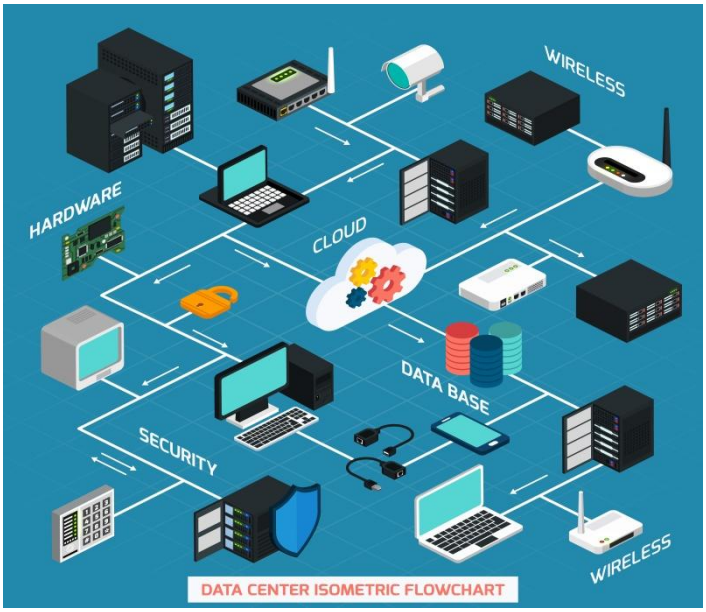
Objective : Study different types of networks , networking devices and various topology

Introduction to Computer Networks :

A computer network is a collection of interconnected computing devices such as computers, servers, printers, and mobile devices that communicate with each other to share data, resources, and services. Computer networks play a crucial role in modern digital communication and are considered the foundation of today's information technology infrastructure.

Computer networks enable various services including email communication, file sharing, video conferencing, cloud computing, online transactions, and internet access. By connecting multiple devices, networks allow users to access shared resources efficiently and reduce the cost of hardware and software usage.

In the early stages, computer networking was limited to simple point-to-point connections between two devices. With the advancement of communication technologies, networks have evolved into large-scale systems capable of connecting millions of devices worldwide. The development of high-speed transmission media, wireless communication, and internet technologies has significantly improved network performance and reliability.



Modern computer networks are designed to be fast, secure, and scalable. They use standard communication protocols to ensure accurate and reliable data transfer. Today, computer networks are widely used in various fields such as education, healthcare, banking, business, and research, making them an essential part of everyday life.

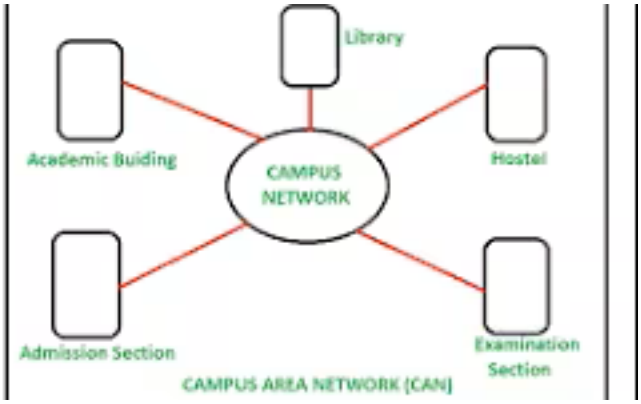
Types of Computer Networks :

Computer networks are classified based on their geographical coverage, purpose, and mode of communication. Each type of network is designed to meet specific communication requirements. The major types of computer networks are explained below in detail.

1.Campus Area Network(CAN) :

A Campus Area Network (CAN) is a computer network that connects multiple Local Area Networks (LANs within a limited geographical area such as a college campus, university, hospital, or corporate campus. It is larger than a LAN but smaller than a Metropolitan Area Network (MAN). CAN is designed to provide high-speed communication between different buildings and departments in a campus.

CAN commonly uses high-speed transmission media such as fiber optic cables, Ethernet, and wireless technologies. It allows centralized management of network resources and provides secure and reliable connectivity for academic and administrative activities.



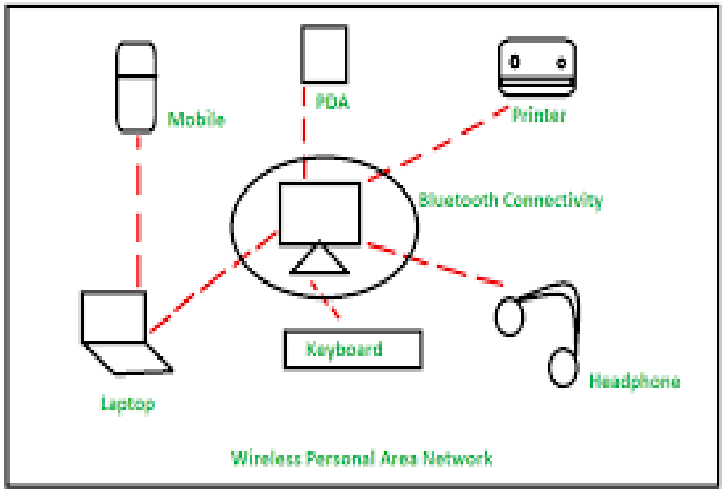
Uses of CAN

- Connecting different departments and buildings in a campus
- Providing internet and intranet services across the campus
- Supporting online learning and digital classrooms
- Sharing servers, printers, and databases
- Enabling communication between students, staff, and administratio

2.Personal Area Network(PAN) :

A Personal Area Network (PAN) is a small computer network that connects personal electronic devices within a very short range, usually up to 10 meters. It is designed to support communication between devices owned and operated by a single user. Devices commonly connected in a PAN include smartphones, laptops, tablets, smartwatches, and wireless headphones.

PAN uses short-range communication technologies such as Bluetooth, Infrared (IR), and ZigBee. These networks are easy to set up, require low power, and are mainly used for personal and temporary connections. Due to its limited range, PAN is suitable only for close-distance communication.



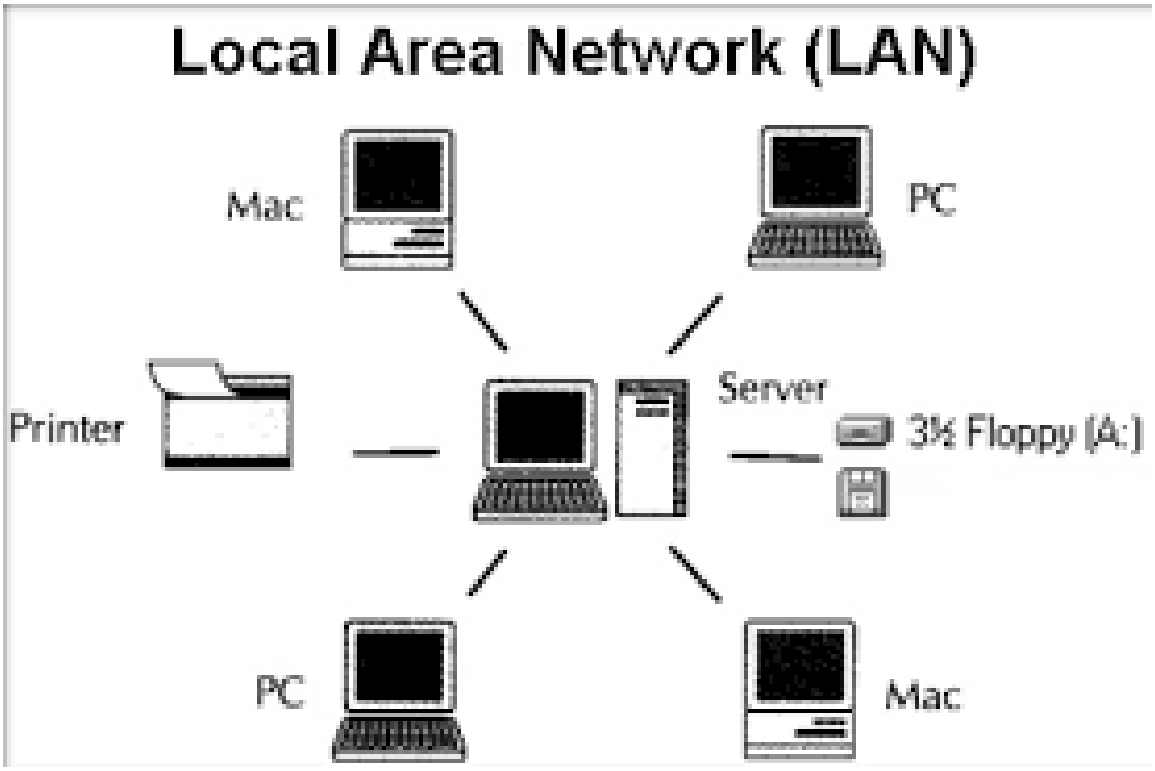
Uses of PAN

- Connecting smartphones with wireless earphones or smartwatches
- File and data transfer between personal devices
- Device synchronization such as phone-to-laptop connection
- Wireless control of personal gadgets
- Connecting peripherals like keyboards, mouse, and printers

3.Local Area Network(LAN) :

A Local Area Network (LAN) is a computer network that connects computers and other devices within a limited geographical area such as a home, office, school, or laboratory. LAN allows devices to communicate with each other at high speed and share resources efficiently. It is generally owned and managed by a single organization or individual.

LAN uses communication technologies such as Ethernet cables and wireless Wi-Fi. Due to its small size, LAN provides reliable connectivity, low delay, and better control over network security.



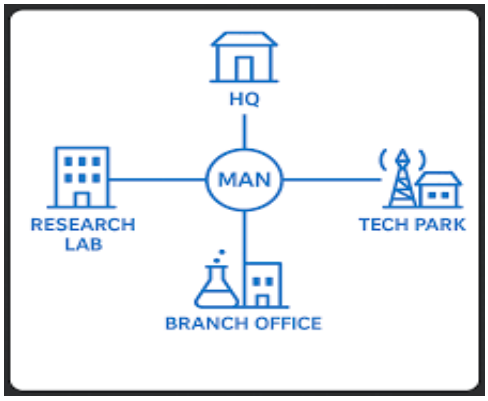
Uses of LAN

- Sharing files and data among computers
- Sharing printers, scanners, and other hardware devices
- Providing internet access within offices and homes
- Supporting internal communication in organizations
- Connecting computers in schools and laboratories

4.Metropolitan Area Network(MAN) :

A Metropolitan Area Network (MAN) is a computer network that covers a larger geographical area than a Local Area Network but smaller than a Wide Area Network. It typically spans a city or a large town and connects multiple LANs within the metropolitan region. MAN is designed to provide high-speed communication between different locations across a city.

MAN uses high-capacity transmission media such as fiber optic cables, microwave links, and leased lines. It is commonly used by universities, government organizations, banks, and service providers to maintain reliable communication across city-wide networks.



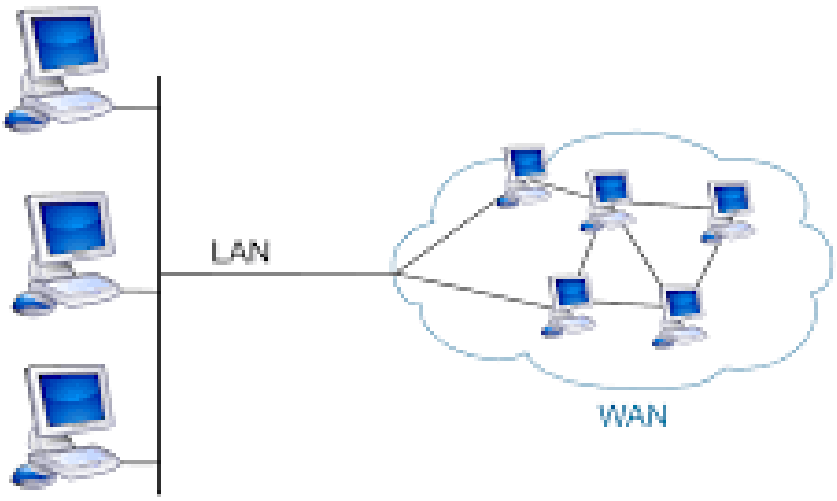
Uses of MAN

- Connecting multiple offices or campuses within a city
- Providing high-speed internet access across metropolitan areas
- Supporting communication for universities and government organizationsEnabling city-wide data sharing and services
- Used in cable television and broadband networks

5.Wide Area Network(WAN) :

A Wide Area Network (WAN) is a computer network that covers a very large geographical area such as countries or continents. WAN connects computers and networks over long distances, allowing communication between remote locations. It is larger than LAN and MAN and is essential for global connectivity.

WAN uses communication technologies such as leased lines, satellites, fiber optic cables, and the Internet. The Internet is the best example of a Wide Area Network. WAN enables long-distance data transmission but usually has higher latency compared to smaller networks.



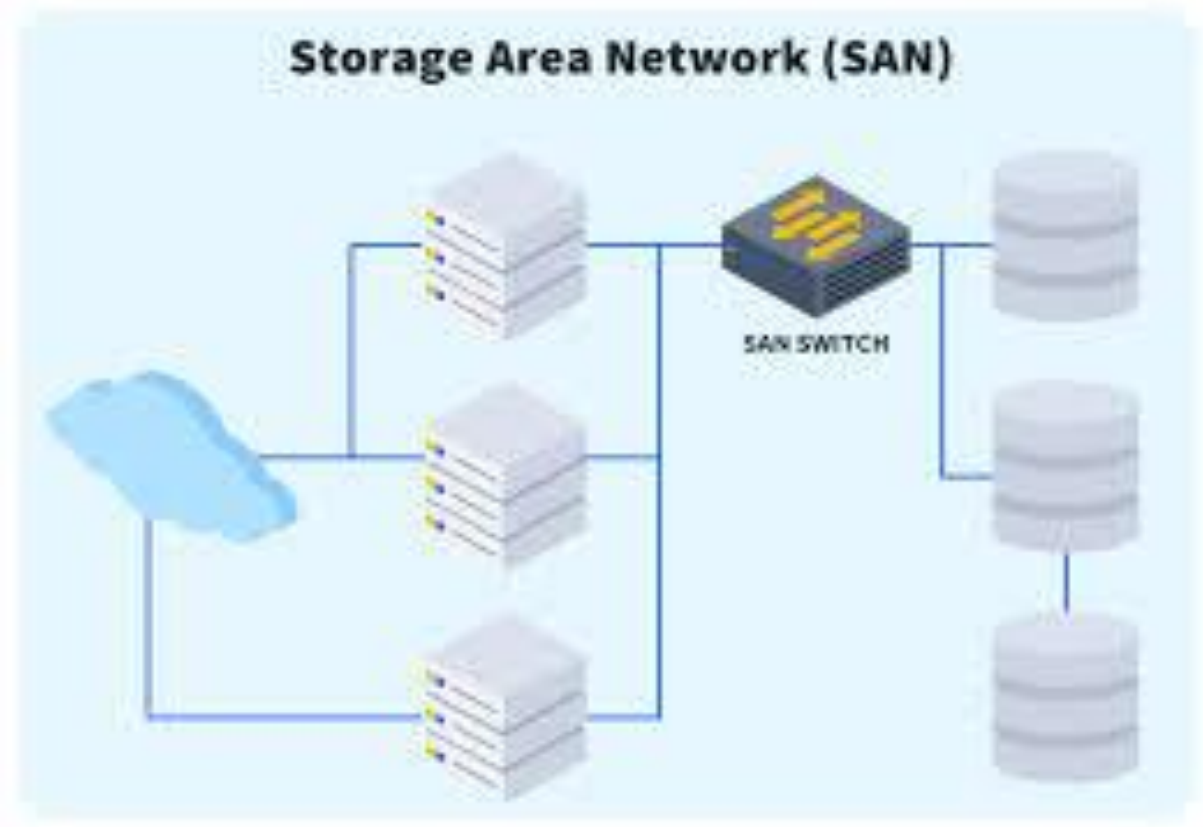
Uses of WAN

- Connecting branch offices located in different cities or countries
- Enabling global communication and data sharing
- Providing internet access across large geographical areas
- Supporting online services such as email, cloud computing, and e-commerce
- Used by multinational companies and service providers

6.Storage Area Network(SAN) :

A Storage Area Network (SAN) is a specialized high-speed network used to connect servers with storage devices such as hard disks and storage arrays. It is mainly used in data centers and large organizations where large amounts of data need to be stored and accessed efficiently. SAN provides a dedicated network for storage, separate from the regular data network.

SAN uses high-speed technologies such as fiber optic cables and specialized protocols. It improves data performance, reliability, and availability by allowing multiple servers to access centralized storage.



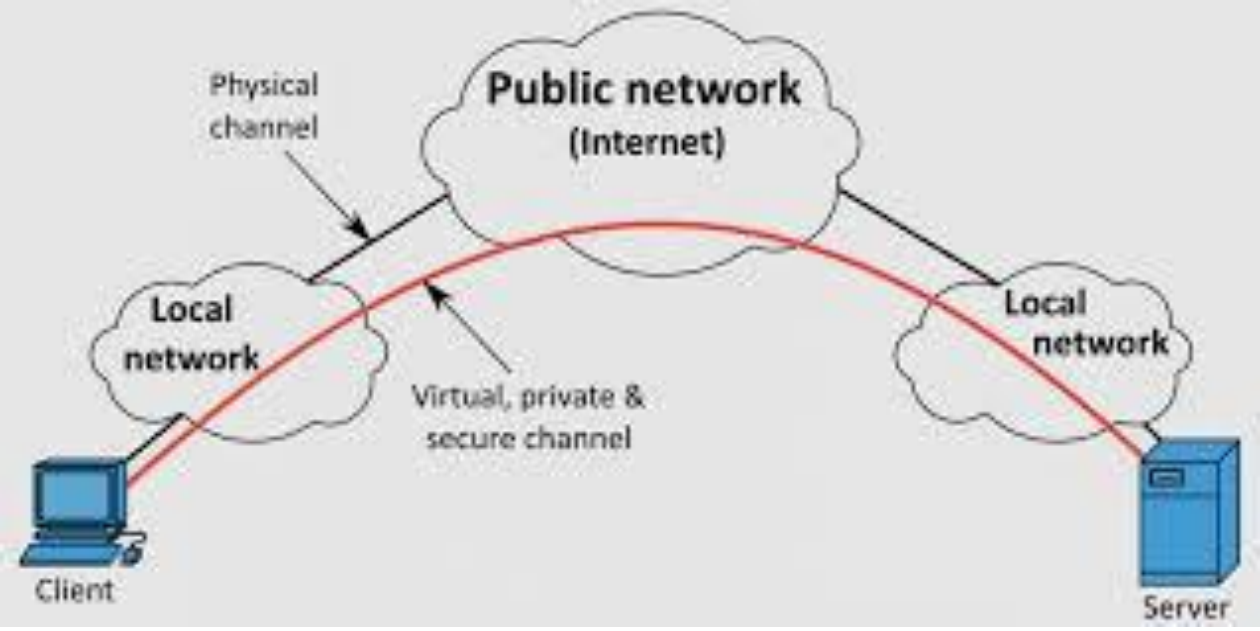
Uses of SAN

- Storing large volumes of organizational data
- Providing fast access to centralized storage
- Supporting data backup and recovery systems
- Used in data centers and cloud environments
- Ensuring high availability and reliability of data

7.Virtual Private Network(VPN) :

A Virtual Private Network (VPN) is a secure private network created over a public network such as the Internet. It allows users to send and receive data securely as if they were directly connected to a private network. VPN uses encryption and tunneling techniques to protect data during transmission.

VPN is commonly used to provide secure remote access to organizational networks. It ensures data privacy and protects information from unauthorized access while communicating over public networks.



Uses of VPN

- Secure remote access to office or campus networks
- Protecting data during internet communication
- Connecting branch offices securely
- Maintaining privacy while using public Wi-Fi
- Used by organizations for secure communication
-

8.Wireless LAN(WLAN) :

A Wireless Local Area Network (WLAN) is a type of Local Area Network that uses wireless communication instead of physical cables to connect devices. It allows devices such as laptops, smartphones, and tablets to connect to the network using radio waves within a limited area like a home, office, school, or campus.

WLAN commonly uses Wi-Fi technology and provides flexibility and mobility to users. Devices can connect to the network without the need for wired connections, making installation easier and more convenient.



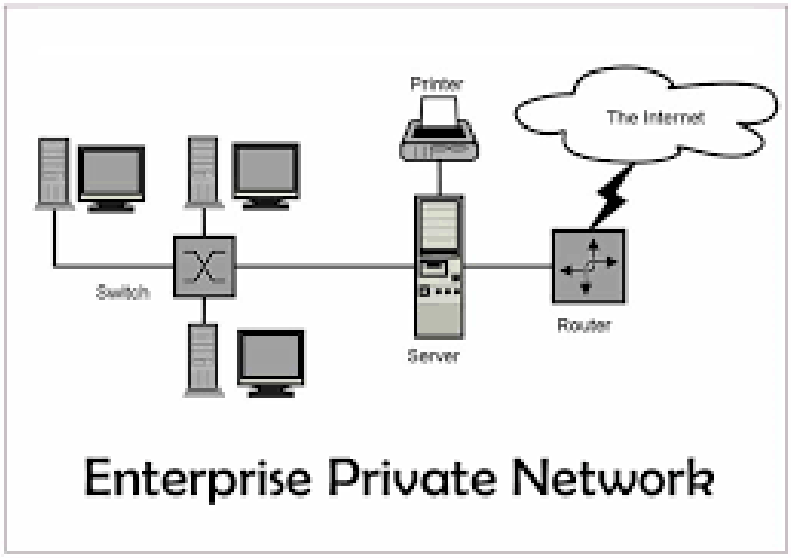
Uses of WLAN

- Providing wireless internet access in homes and offices
- Connecting mobile devices such as laptops and smartphones
- Used in schools, colleges, and campuses
- Supporting public Wi-Fi in airports, cafés, and hotels
- Enabling flexible and portable network access

9.Enterprise Private Network(EPN) :

An Enterprise Private Network (EPS) is a private network built and maintained by an organization to securely connect its offices, departments, and branches. It is not accessible to the public and provides controlled access to authorized employees only. EPS allows efficient communication and data sharing within the organization.

EPS uses high-speed wired connections like fiber optics, leased lines, and sometimes secure wireless links. It ensures reliable, secure, and centralized management of organizational resources.



Uses of EPS

- Connecting multiple branches of a company or organization
- Secure communication between employees and departments
- Centralized access to servers, databases, and applications
- Supporting internal collaboration and resource sharing
- Ensuring secure and private organizational communication

Netwroking Devices :

Networking devices are hardware components that help computers, servers, and other devices communicate with each other within a network. They play a crucial role in managing data flow, connecting networks, and maintaining security and performance. Different devices operate at different layers of the network (Physical, Data Link, Network, etc.) and have specific purposes.

The main networking devices are:

Physical Layer Devices(Layer-1) :

Physical layer devices operate at the first layer (Physical Layer) of the OSI model. They are responsible for the transmission and reception of raw data bits over physical media such as cables or wireless signals. Common physical layer devices include Hub, Repeater, Modem, Cables, and Connectors.

1. Hub

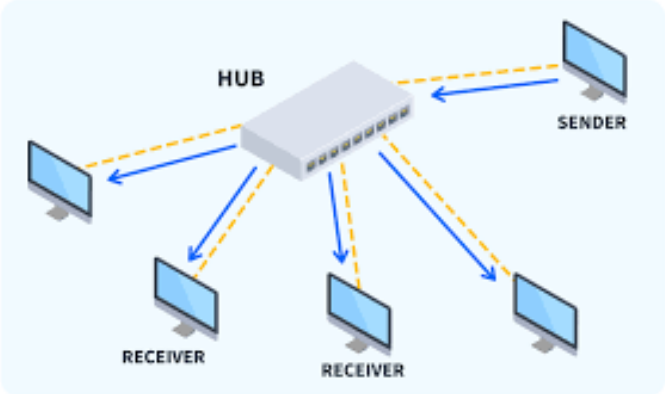
A hub is a basic networking device that connects multiple computers or devices in a LAN. It works at the physical layer and forwards data received from one port to all other ports without filtering.

Uses:

- Connecting multiple devices in a small LAN
- Basic network communication in homes or labs

Pros:

- Simple and inexpensive
- Easy to set up

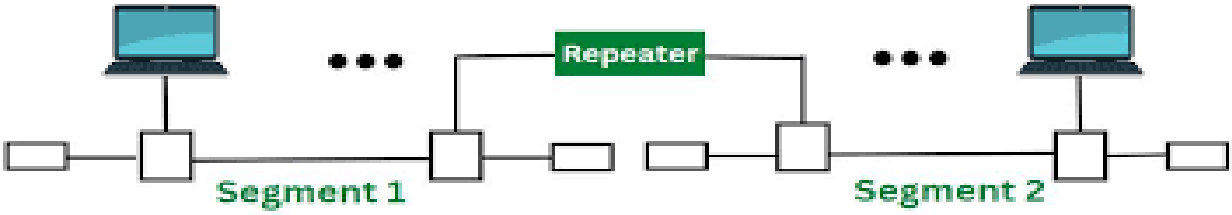


Cons:

- Sends data to all devices → inefficient
- Less secure (data can be intercepted by any connected device)

2. Repeater

A repeater is a device used to regenerate and amplify signals to extend the distance of a network. It works at the physical layer by boosting weak or degraded signals.



Uses:

- Extending the coverage of a LAN or WAN
- Boosting signal strength over long cables

Pros:

- Simple device, cost-effective
- Extends network range

Cons:

- Does not filter or control traffic
- Only works with same type of network signals

3. Modem

A modem (modulator-demodulator) converts digital signals from a computer into analog signals for transmission over telephone or cable lines and vice versa.

Uses:

- Connecting home or office networks to Internet service providers
- Enabling data transmission over analog lines

Pros:

- Provides Internet connectivity
- Compatible with standard telephone lines



Cons:

- Speed is limited compared to modern broadband
- Requires proper configuration

4.Cables

Cables are physical media used to transmit data signals between devices. Common types include Twisted Pair, Coaxial, and Fiber Optic cables.

Uses:

- Connecting computers, hubs, switches, and routers
- Providing wired network connections for LANs



Pros:

- Reliable and stable connections
- High data transmission speed for short distances (especially fiber optics)

Cons:

- Installation can be cumbersome
- Limited mobility for connected devices

5. Connectors

Connectors are devices used to join cables to networking devices or extend cables. Examples include RJ-45, RJ-11, SC, ST connectors.

Uses:

- Connecting cables to computers, switches, routers, and hubs
- Ensuring stable and secure connections



Pros:

- Easy to install
- Provides secure and reliable physical connections

Cons:

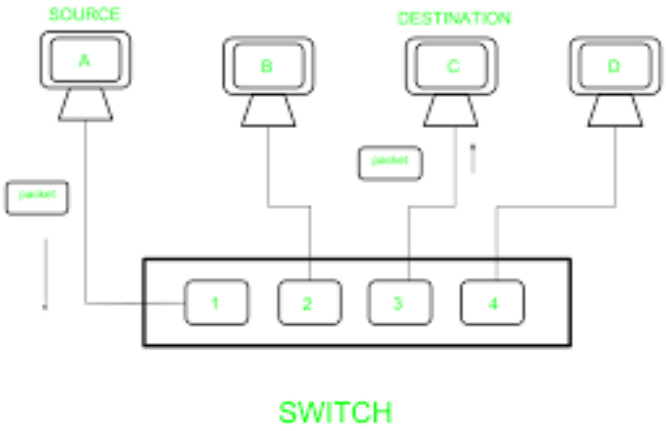
- Poor quality connectors can reduce network performance
- Can wear out with repeated use

DataLink Layer Devices(Layer-2) :

Data Link Layer devices operate at the second layer of the OSI model. Their main function is to control how data is transferred between devices on the same network and ensure reliable and error-free communication. Common data link layer devices include Switch, Bridge, and Network Interface Card (NIC).

1.Switch

A switch is a networking device that connects multiple devices in a LAN and works at the data link layer. It forwards data only to the intended device using MAC addresses, unlike a hub which sends data to all devices.



SWITCH

Uses:

- Connecting multiple computers in offices, schools, and labs
- Managing traffic efficiently in a LAN
- Reducing network collisions

Pros:

- Efficient and fast data transfer
- Reduces network congestion
- More secure than a hub

Cons:

- More expensive than a hub
- Requires proper configuration for large networks

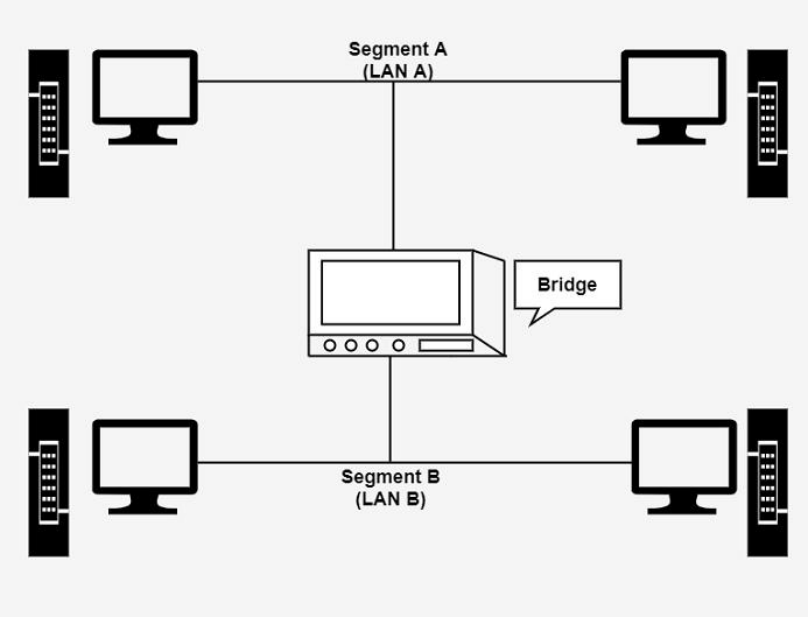
2.Bridge

A bridge connects two or more LAN segments and filters traffic to reduce congestion. It operates at the data link layer and uses MAC addresses to decide whether to forward or block data.

Uses:

- Connecting separate LAN segments into a single network
- Reducing unnecessary traffic between network segments
- Extending LAN coverage

:



pros

- Reduces network collisions and congestion
- Efficient in segmenting networks

Cons:

- Limited to connecting LANs only
- Slower than switches in large networks

3. Network Interface Card (NIC)

A Network Interface Card (NIC) is a hardware component installed in a computer or device to allow it to connect to a network. It can be wired (Ethernet NIC) or wireless (Wi-Fi NIC). NIC operates at the data link layer and converts data into a format suitable for transmission over the network.

Uses:

- Connecting computers and devices to LANs or WLANs
- Enabling data communication over networks
- Providing unique MAC addresses for each device

Pros:

- Essential for network connectivity
- Easy to install and replace
- Supports wired and wireless connections

Cons:

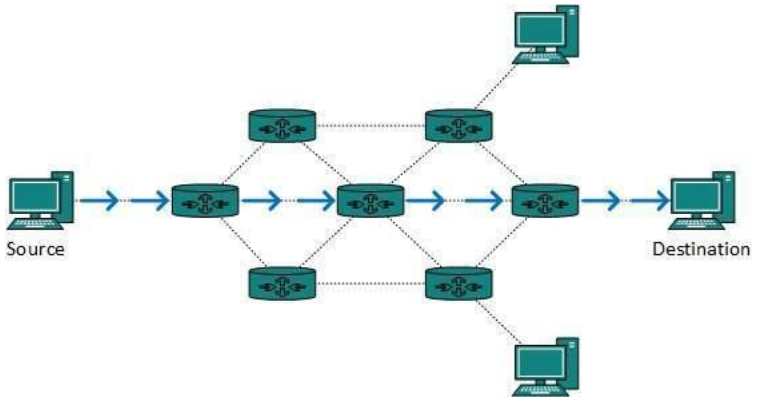
- Hardware failure can disconnect the device from the network
- Requires compatible drivers and proper configuration

Network Layer Devices(Layer-3) :

Network Layer devices operate at the third layer of the OSI model. Their main function is to forward data packets between different networks based on logical addressing (IP addresses). Common Network Layer devices include Router and Brouter (Bridge Router).

1.Router

A router is a device that connects two or more networks and directs data packets from the source network to the destination network using IP addresses. It operates at the network layer and can connect LANs to WANs or the Internet.



Uses:

- Connecting home or office networks to the Internet
- Routing data packets between different networks
- Managing traffic in large networks
- Supporting network security and NAT (Network Address Translation)

Pros:

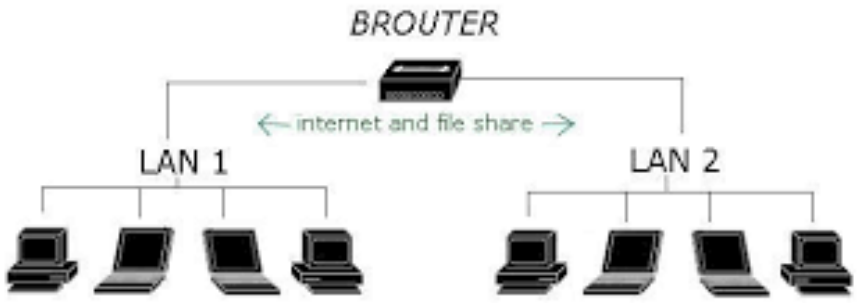
- Enables communication between different networks
- Provides Internet connectivity
- Supports network security features like firewalls and NAT

Cons:

- More expensive than switches or hubs
- Requires proper configuration and management
- Slower than switches for local data transfer

2.Brouter(BridgeRouter)

A Brouter is a combination of a bridge and a router. It can forward packets like a router between networks and filter traffic like a bridge within a network segment. Brouters operate at both data link layer and network layer.



Uses:

- Connecting LANs with different protocols
- Filtering and forwarding packets within and between networks
- Reducing congestion in large networks

Pros:

- Combines features of bridge and router
- Efficient in handling multiple types of network traffic
- Reduces unnecessary network traffic

Cons:

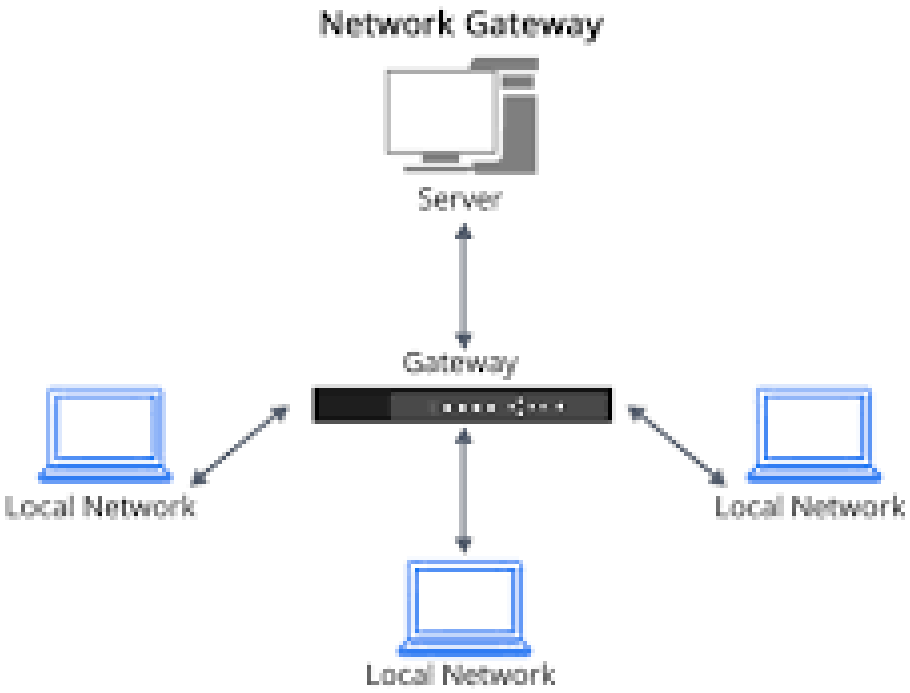
- Expensive and complex
- Requires technical expertise to configure
- Slower than dedicated devices for high-traffic networks
-

Higher Layer and Specialized Devices(Layer 4-7) :

Devices operating at Layers 4 to 7 of the OSI model provide advanced functions like security, protocol conversion, traffic management, and application-level services. Common higher layer devices include Gateway and Firewall.

1.Gateway

A gateway is a networking device that connects two different networks that may use different protocols. It operates at the higher layers (Layer 4–7) and can perform protocol conversion, data translation, and routing functions to allow communication between incompatible networks.



Uses:

- Connecting enterprise networks to the Internet
- Translating between different network protocols
- Enabling communication between dissimilar systems (e.g., LAN to mainframe)
- Integrating cloud services with internal networks

Pros:

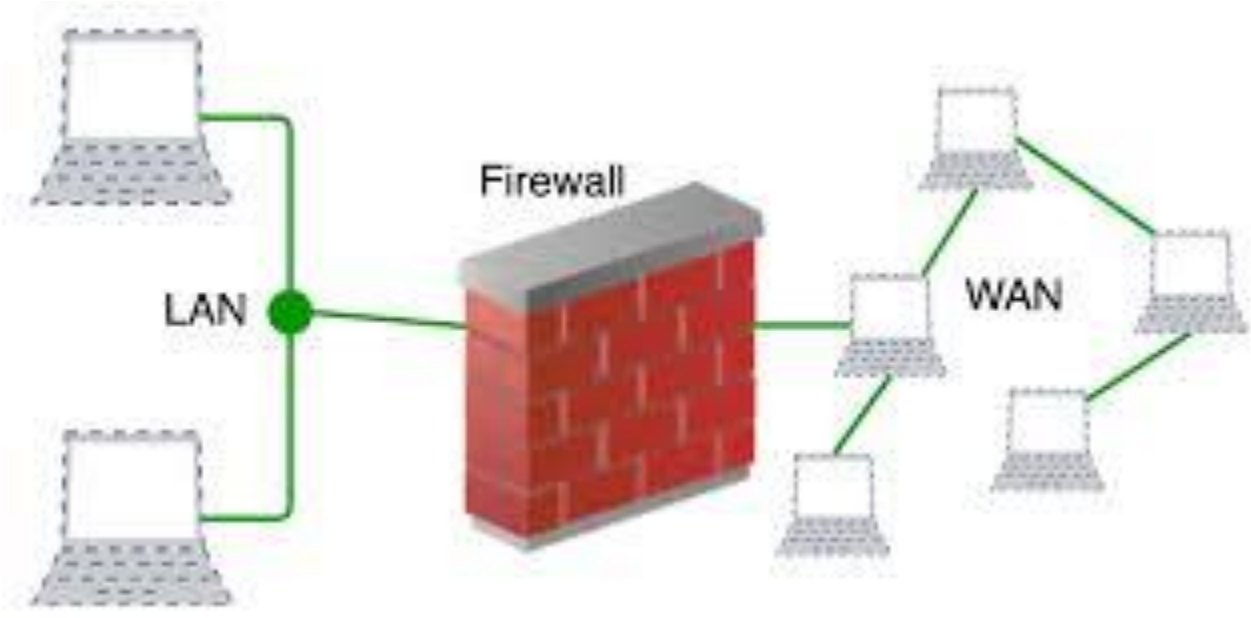
- Allows interoperability between different networks
- Supports multiple protocols and applications
- Enables secure and controlled communication

Cons:

- Complex and expensive
- Requires skilled configuration and management
- Can become a bottleneck if overloaded

2.Firewall

A firewall is a network security device that monitors and controls incoming and outgoing network traffic based on predefined security rules. It operates at layers 4–7 and is used to protect networks from unauthorized access, malware, and cyber attacks.



Uses:

- Protecting internal networks from external threats
- Filtering traffic based on IP addresses, ports, and protocols
- Preventing unauthorized access to sensitive data
- Implementing security policies in organizations

Pros:

- Enhances network security
- Provides control over network traffic
- Prevents cyber attacks and data breaches

Cons:

- Can slow down network performance if misconfigured
- Complex to manage in large networks
- Does not protect against internal threats

Network topologies :

Network topology refers to the physical or logical arrangement of devices such as computers, servers, switches, and routers in a computer network. It determines how devices are connected, how data travels between them, and how network resources are shared.

The topology of a network is a critical factor in determining the performance, reliability, and scalability of the network. It affects the speed of data transmission, ease of troubleshooting, fault tolerance, and cost of installation. Choosing the correct topology ensures that the network can handle the intended number of devices, traffic load, and future expansion.

Network topologies can be classified into two types:

1. Physical Topology: The actual physical layout of the cables, devices, and connections in the network. For example, in a star topology, all devices are physically connected to a central switch.
2. Logical Topology: The path along which data flows within the network, which may differ from the physical layout. For example, data in a logical ring can flow in a loop even if devices are physically arranged in a star.

Proper design of network topology also impacts network security and maintenance. Some topologies, like mesh, provide high fault tolerance but are expensive, while simpler topologies like bus are cheap but more prone to failure. Network topology decisions are therefore based on the size of the network, the number of devices, budget, and required reliability.

By understanding different network topologies, network designers can maximize performance, minimize downtime, and ensure efficient resource sharing.

Types of Network Topologies :

Network topology refers to the physical or logical arrangement of devices in a network and defines how data flows between them. The topology of a network affects its performance, reliability, scalability, and cost. The most common topologies are Bus, Star, Ring, Mesh, and Hybrid.

1. Bus Topology

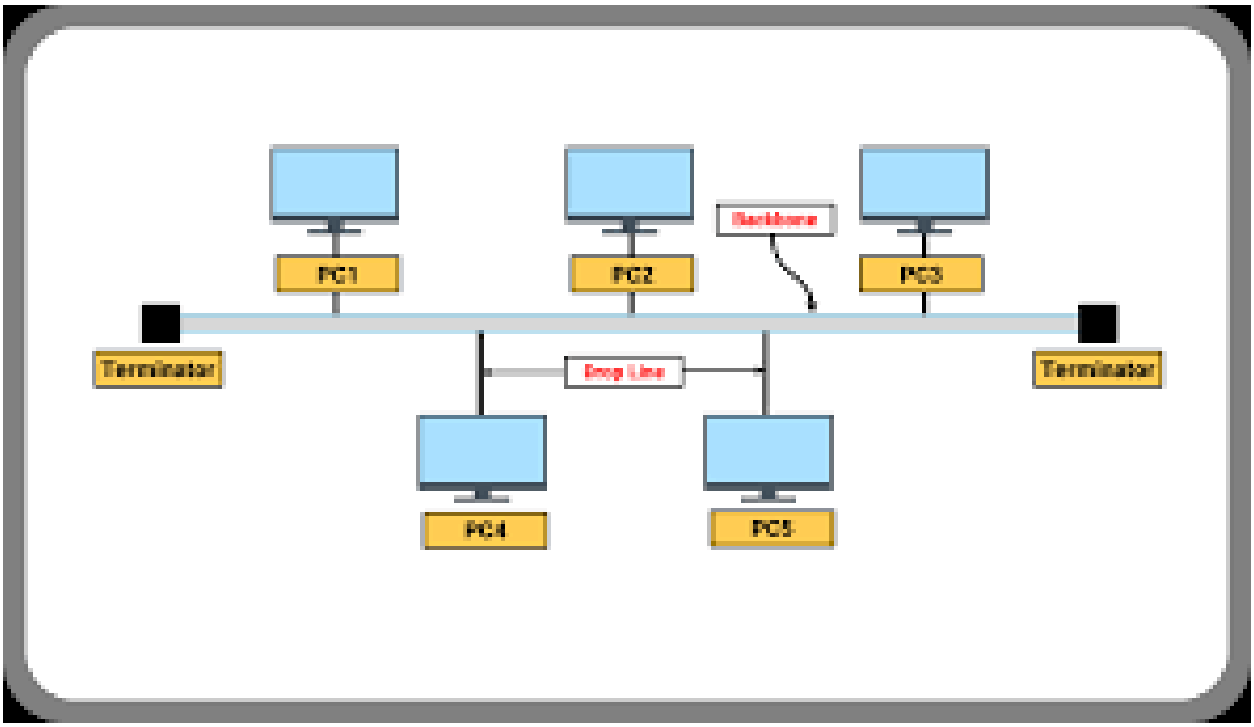
A Bus topology is a network in which all devices are connected to a single central communication line (backbone). Data travels along the backbone in both directions and is received by all devices, but only the intended recipient accepts it.

Components:

- Single backbone cable (coaxial or twisted pair)
- Terminators at both ends of the backbone
- Computers and devices connected via T-connectors

Working:

When a device sends data, it travels along the ball devices. Only the device whose address matches the data accepts and processes it; the rest ignore it.



Pros:

- Simple and easy to implement
- Requires less cable compared to other topologies
- Cost-effective for small networks

Cons:

- A break in the backbone can take down the entire network
- Difficult to troubleshoot faults
- Not suitable for large networks due to data collisions

Real-World Application:

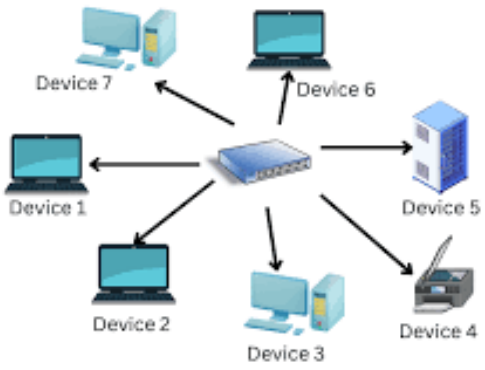
- Small offices and laboratories
- Early Ethernet networks (10Base2)

2. Star Topology

In Star topology, all devices are connected to a central device such as a switch or hub. The central device acts as a mediator for communication between devices.

Components:

- Central hub, switch, or router
- Devices (computers, printers)
- Connecting cables (twisted pair, fiber optic)



Working:

Data from a device is sent to the central device, which then forwards it to the destination device. Communication is controlled by the central hub or switch.

Pros:

- Easy to manage and troubleshoot
- Failure of one device does not affect others
- Easy to add or remove devices

Cons:

- Central device failure can disrupt the entire network
- Requires more cabling than bus topology
- Slightly higher installation cost

Real-World Application:

- Modern office networks
- Schools and colleges
- Home Wi-Fi networks

3. Ring Topology

In Ring topology, each device is connected to exactly **two other devices**, forming a closed loop. Data travels in a **single or dual direction** along the ring.

Components:

- Devices connected in a circular path
- Network interface cards (NIC)
- Optional repeaters for signal regeneration

Working:

Data is sent from one device to the next around the ring until it reaches the intended recipient. In a dual-ring topology, data can travel in both directions for redundancy.

Pros:

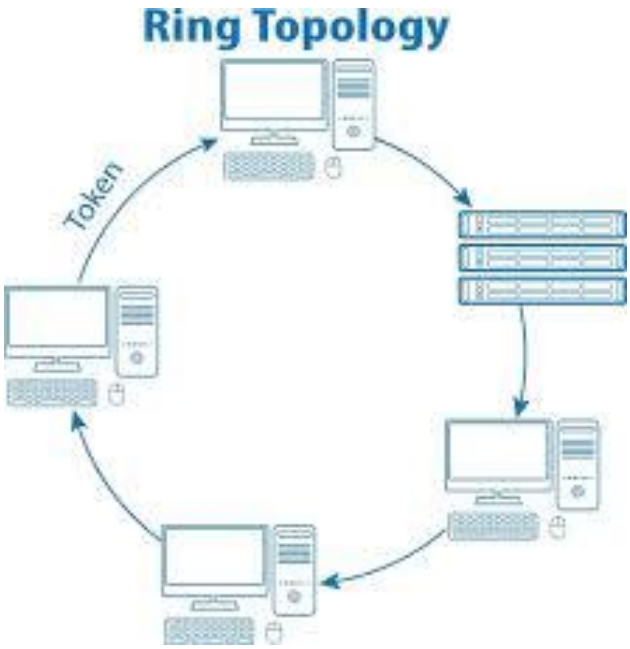
- Organized data flow reduces collisions
- Works well for medium-sized networks
- Predictable performance

Cons:

- Failure of one device or link can disrupt the network (unless dual ring is used)
- Difficult to install and reconfigure
- Adding or removing devices requires temporary network shutdown

Real-World Application:

- Token Ring networks
- Some college and corporate LANs in the past



4.Mesh Topology

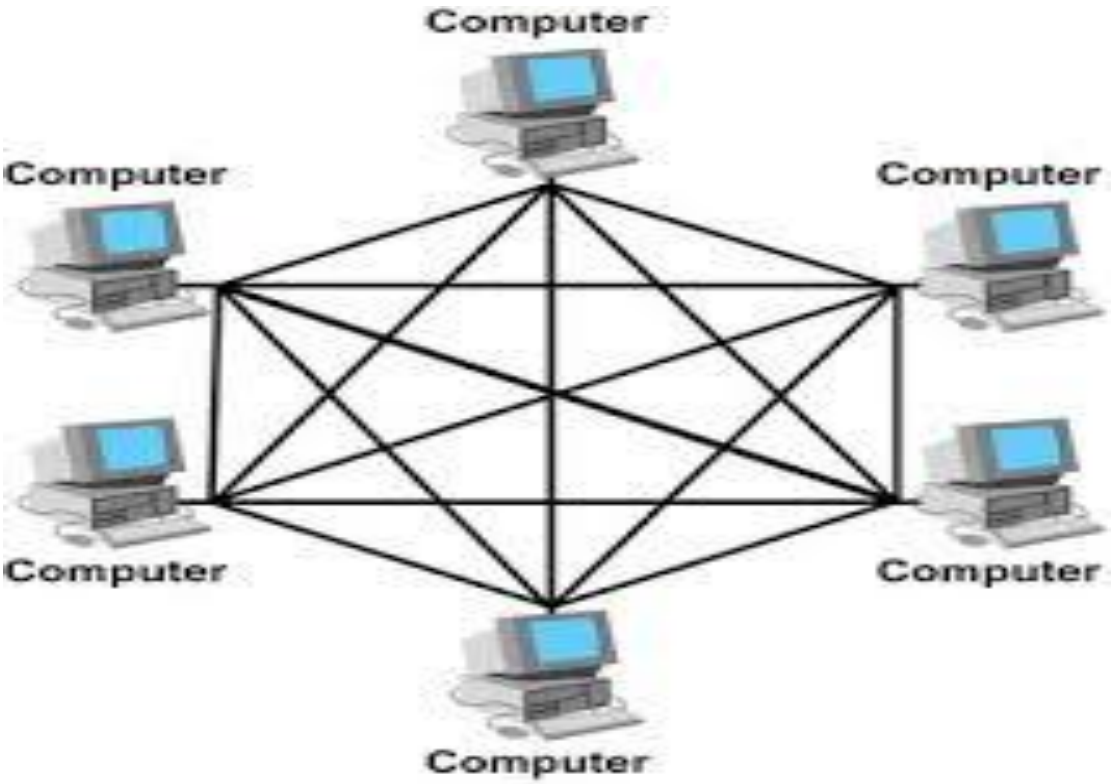
In Mesh topology, every device is connected to **every other device** in the network, creating multiple paths for data.

Components**:**

- Devices (computers, routers, switches)
- Multiple interconnecting cables or wireless links

Working:

Data can travel along multiple paths from source to destination. If one path fails, an alternate path is used, ensuring high reliability.



Pros:

- Highly reliable and fault-tolerant
- No single point of failure
- Redundant paths improve network performance

Cons:

- Expensive due to large number of connections
- Complex installation and maintenance
- Requires more cabling and management

Real-World Application:

- Mission-critical networks like banks, military, and hospitals
- WAN backbone networks
- Large corporate networks

5. Hybrid Topology

Hybrid topology is a combination of two or more topologies, such as star-bus or star-ring, designed to leverage the advantages of each.

Components:

- Devices, central hubs/switches, backbone cables
- Combination of topology components (e.g., star hubs + bus backbone)

Working:
Different parts of the network use different topologies. For example, multiple star networks may be connected via a bus or ring backbone. Communication depends on the arrangement of sub-topologies.

Pros:

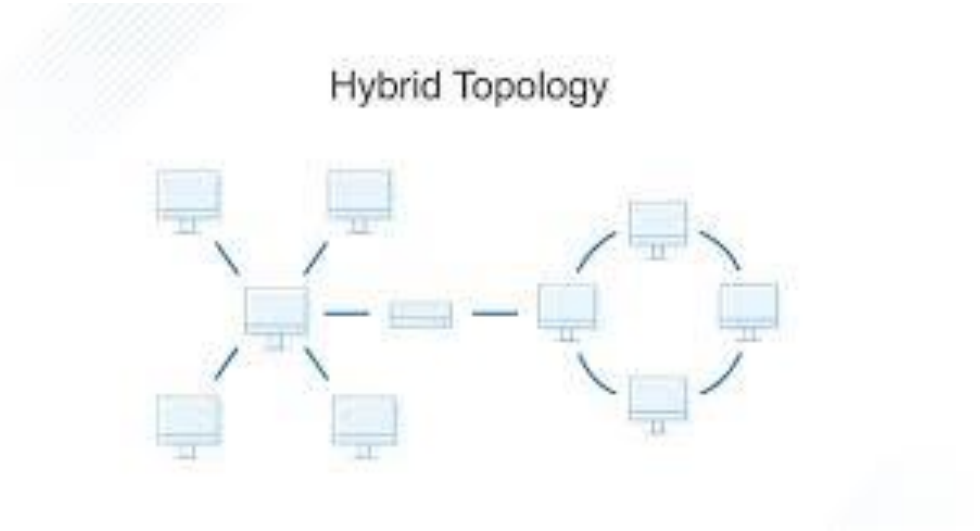
- Flexible and scalable
- Combines advantages of multiple topologies
- Can handle large networks efficiently

Cons:

- Complex to design and maintain
- Higher installation cost
- Troubleshooting can be difficult

Real-World Application:

- Large corporate networks with multiple departments
- University campuses
- Enterprise networks combining different buildings or branches



Experiment – 2:

Objective : Study of Various Types of Guided and Unguided Transmission Media

Transmission Media :

In computer networks, transmission media refers to the physical or logical path used to transmit data from one device to another**.** The choice of transmission media is crucial because it affects network speed, distance, reliability, cost, and security**.**

Transmission media are broadly classified into two categories:

1. **Guided(Wired)Media:**
   Data is transmitted through a physical medium, such as cables. The signal is guided along a solid path. Examples include twisted pair cables, coaxial cables, and fiber optic cables.
2. **Unguided(Wireless)Media:**
   Data is transmitted without physical connections, using electromagnetic waves such as radio waves, microwaves, infrared, or satellite signals. Wireless media are used where mobility and long-distance coverage are important.

The choice of media depends on factors like bandwidth requirements, distance between devices, installation cost, environmental conditions, and security needs.

Types of Transmission Media

A. Guided (Wired) Media

1.Twisted Pair Cable

Twisted pair cable consists of pairs of insulated copper wires twisted together. Twisting reduces electromagnetic interference from external sources and from adjacent pairs.

Components:

- Copper conductor (usually 24 AWG)
- Insulation layer

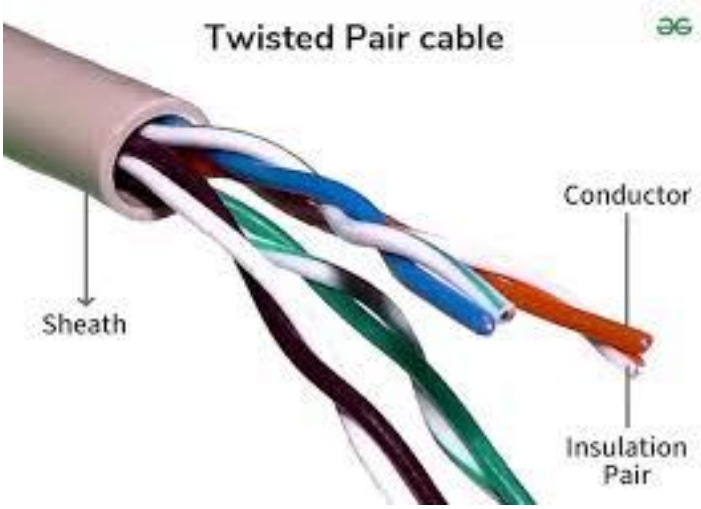
- Twisting pairs to reduce interference
- Outer protective jacket

Working:

Data signals are sent as electrical pulses along the copper wires. Twisting the pairs reduces crosstalk and interference, ensuring reliable communication.

Uses:

- Local Area Networks (LAN)
- Telephone networks
- Home networking



Pros:

- Low cost
- Flexible and easy to install
- Widely available

Cons:

- Limited bandwidth (usually up to 100 Mbps for UTP)
- Susceptible to electromagnetic interference
- Limited distance (usually <100 meters)

Real-World Application:

- Office LAN connections
- Telephone communication
- Early Ethernet networks

2. Coaxial Cable

A coaxial cable has a central conductor surrounded by insulation, a metallic shield, and an outer protective jacket. This design allows it to carry high-frequency signals with minimal interference.

Components:

- Central copper conductor
- Dielectric insulator
- Metallic shield (braid or foil)
- Outer jacket

Working:
Data signals are transmitted as electrical pulses through the central conductor. The shield prevents interference from external signals.

Uses:

- Cable TV networks
- Broadband Internet
- Older Ethernet LANs



Pros:

- Higher bandwidth than twisted pair
- Resistant to electromagnetic interference
- Reliable for medium distances

Cons:

- More expensive than twisted pair
- Less flexible
- Requires connectors and proper installation

Real-World Application:

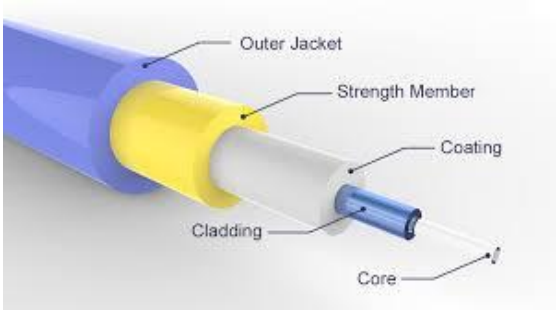
- Television cable networks
- Broadband Internet connections
- Connecting network devices in offices

3. Fiber Optic Cable

Fiber optic cables transmit data as light pulses through glass or plastic fibers, offering extremely high-speed communication over long distances.

Components:

- Core (glass/plastic for light transmission)
- Cladding (reflects light back into the core)
- Protective coating
- Outer jacket



Working:
Data is converted into light signals using a laser or LED. The light travels through the core and is reflected by the cladding. At the receiving end, it is converted back into electrical signals.

Uses:

- Internet backbone
- High-speed LANs and WANs
- Undersea cables for global communication

Pros:

- Very high bandwidth
- Immune to electromagnetic interference
- Can transmit over long distances

Cons:

- Expensive installation
- Fragile and requires careful handling
- Needs specialized equipment

Real-World Application:

- Global Internet backbone
- High-speed corporate networks
- Telecommunication networks

B. Unguided (Wireless) Media

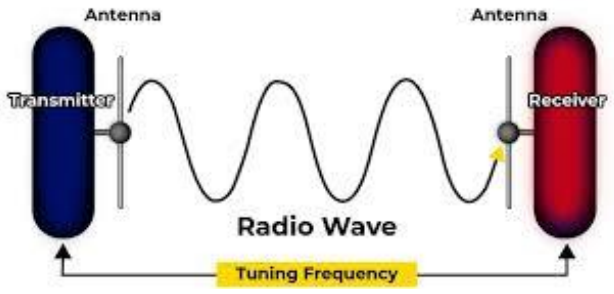
1.Radio Waves

Radio waves are electromagnetic waves used to transmit data wirelessly over short to long distances.

Working:
Data is modulated onto radio frequency signals and transmitted through the air. Receivers demodulate the signal to extract the data.

Uses:

- Wi-Fi networks
- AM/FM radio
- Mobile communication



Pros:

- Wide coverage
- Supports mobility
- Easy to install

Cons:

- Susceptible to interference
- Security risks
- Limited bandwidth

Real-World Application:

- Home Wi-Fi
- Radio broadcasting
- Mobile networks

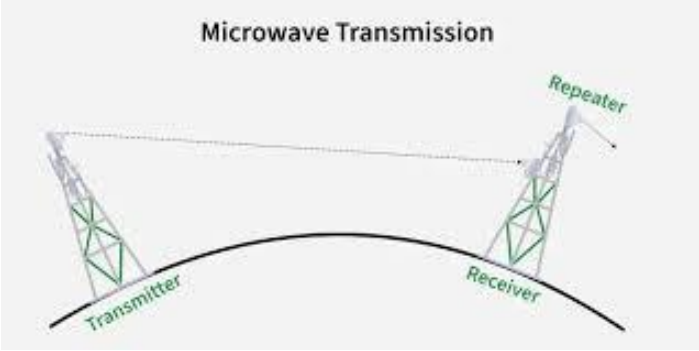
2. Microwave

Microwave transmission uses high-frequency radio waves in the range of GHz. It usually requires line-of-sight between transmitting and receiving antennas.

Working:
Data is converted into microwave signals and transmitted from one antenna to another. Repeaters or relay towers are used for long distances.

Uses:

- Long-distance communication
- Satellite communication
- Point-to-point networks



Pros:

- High-speed data transfer
- Covers long distances
- Reliable for line-of-sight communication

Cons:

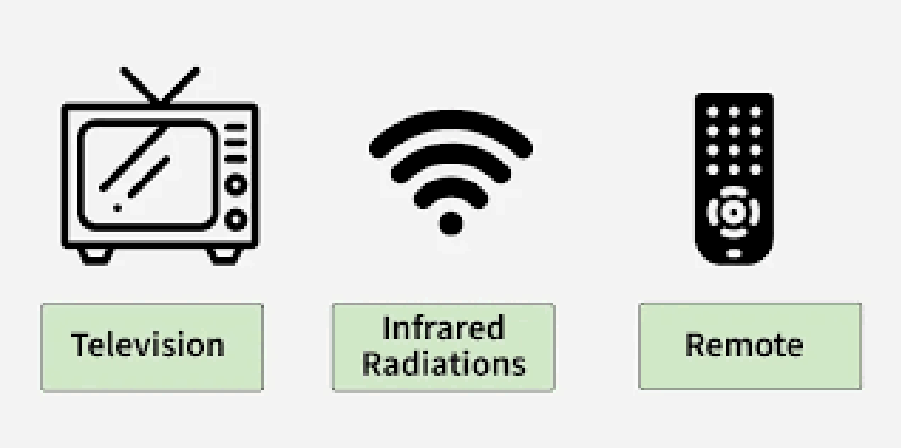
- Requires clear line-of-sight
- Expensive equipment
- Weather can affect signals

Real-World Application:

- Satellite uplink/downlink
- Television networks
- Telephone trunk lines

3. Infrared (IR)

Infrared waves are used for short-range communication. They transmit data as light pulses, typically in devices within a room.



Working**:**
Data is sent as infrared light. Receivers detect and decode the pulses back into data.

Uses:

- Remote controls
- Short-range wireless communication between devices
- IR sensors

Pros:

- Secure (signals do not pass through walls)
- Simple and low-cost
- No interference from radio signals

Cons:

- Very short range
- Cannot penetrate obstacles
- Requires direct line-of-sight

Real-World Application:

- TV remotes
- Wireless keyboards/mice
- IR sensors in devices

4. Satellite Communication

Satellites relay data over long distances**,** using radio or microwave signals, to provide global coverage.

Working:
Data is sent from an Earth station to a satellite (uplink), which then retransmits it back to another Earth station (downlink).

Uses:

- GPS systems
- Global TV broadcasting
- Internet and telecommunication

Pros:

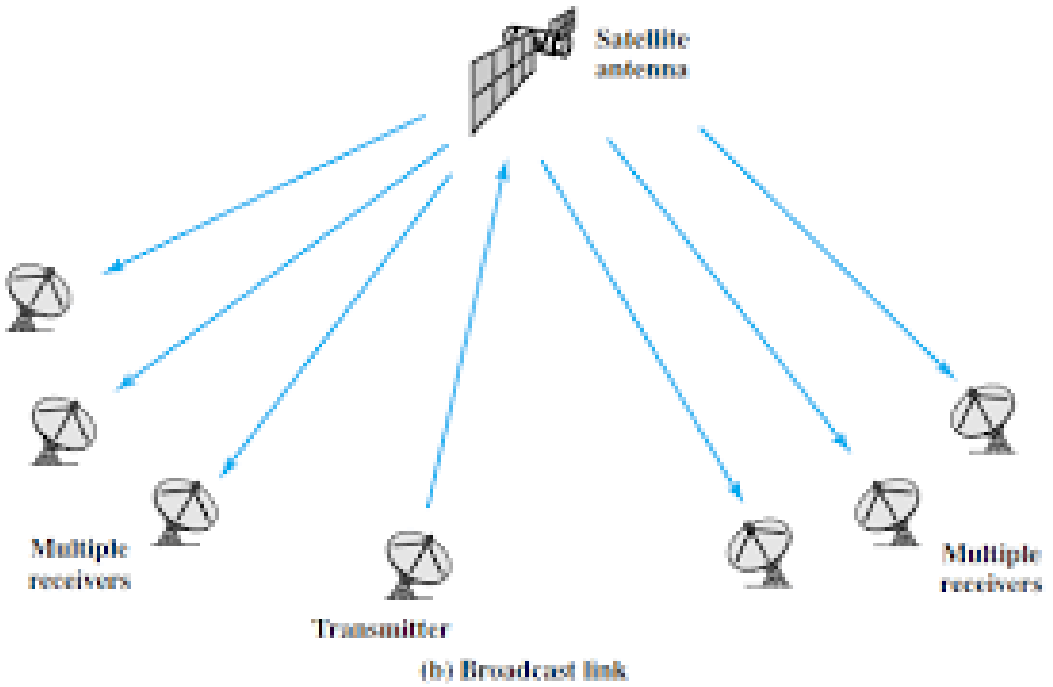
- Global coverage
- Can connect remote areas
- Supports long-distance communication

Cons:

- Expensive to deploy and maintain
- High latency
- Weather conditions may affect signals

Real-World Application**:**

- Satellite TV
- Global Internet access (VSAT)
- GPS navigation



(b) Broadcast link